

SAI RoloTech - Claude Code Operating Guide

Complete Reference Manual v1.0



TABLE OF CONTENTS

1. [How I Work](#)
 2. [Pre-Task Verification](#)
 3. [Project Structure](#)
 4. [Services Status](#)
 5. [Workflow Rules](#)
 6. [Skills System](#)
 7. [Testing Protocol](#)
 8. [API Keys](#)
 9. [Commands Reference](#)
 10. [Play Store Guide](#)
-



HOW I WORK

My Identity

- **Name:** Claude Code (Opus 4.6)
- **Creator:** Anthropic
- **Type:** AI Coding Assistant
- **Specialty:** Full-stack development, automation, deployment

My Approach

1. READ RULES FIRST → Understand workflow
2. VERIFY SERVICES → Test before starting
3. EXECUTE TASK → Code, test, verify
4. UPDATE MEMORY → Save progress
5. COMMIT → Save to git

Key Principles

- ☒ Always verify services before any task
 - ☒ Test code before claiming "done"
 - ☒ Follow rule book strictly
 - ☒ Update memory after every task
 - ☒ Use cheapest working model (Gemini > OpenRouter > Claude)
-



PRE-TASK VERIFICATION

MUST DO BEFORE ANY TASK:

PRE-TASK CHECKLIST

1. cat .claude/memory.json → Load memory
2. cat RULES.md → Read rules
3. cat .claude/CHECKPOINT.md → Check checkpoint
4. Test services → Verify before starting
5. Read project files → Understand structure

Service Verification Commands:

```
# Gemini API Test
curl -X POST "https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-
flash:generateContent?key=YOUR_KEY" \
-H "Content-Type: application/json" \
-d '{"contents":[{"parts":[{"text":"test"}]}]}'

# If 403 Error → API Key Blocked → Get new key

# Backend Server Test
curl http://localhost:5000/api/health

# If fail → Start server: npm run server

# Git Status
git status
```

PROJECT STRUCTURE

Main Repository

```
cloud-code-extension/
├─ crm/                # 3D Industrial CRM (crm/index.html)
├─ crm-official/       # Official CRM (React + Express)
│  └─ mobile/          # Expo React Native app
│  └─ server/          # Express backend
│  └─ src/              # Frontend pages & components
│     └─ android-release.aab # Built Android app
├─ sai-rolotech-engine/ # AI Design Engine
├─ skills/             # Downloaded skills
└─ CLAUDE.md           # Main guide
```

Key Files

File	Purpose
CLAUDE.md	Main project instructions
RULES.md	Workflow rules & commands
.claude/memory.json	Session memory
.claude/CHECKPOINT.md	Validation key

SERVICES STATUS

Current Status (2026-04-14)

SERVICES STATUS
✓ WORKING: └ Local CRM (crm/index.html) - http://localhost:3000 └ Backend Server - Port 5000 (mock mode) └ Git Repository - Committed to GitHub ✗ ISSUES: └ Gemini API - BLOCKED (403 - Leaked key reported) └ EAS Account - Wrong token (vipu1164) └ api.sairolotech.com - NOT REACHABLE ⚠ NEEDS ATTENTION: └ New Gemini API key required └ Correct Expo token needed for EAS └ AAB upload to Play Store

WORKFLOW RULES

Core Workflow

MANDATORY WORKFLOW
1. READ RULES → cat RULES.md (First!) 2. VERIFY → Test services before starting 3. EXECUTE → Code, test, show output 4. UPDATE → Save to memory.json

```
5. COMMIT → Every 5 steps, git commit + push
```

Checkpoint System

Every 5 Steps:

```
git add . && git commit -m "checkpoint: [what was done]" && git push
```

Checkpoints:

- Session start → Read RULES.md
- After coding → Test immediately
- After testing → Show output
- After completion → Commit + Push
- Before new task → Read RULES.md again

Testing Rules

- ❌ WRONG: "Ho gaya, code likh diya"
- ✅ RIGHT: "Code likha, test run kiya, output dekho..."

Test Commands:

```
npm run dev / build / test
node src/index.ts
npx tsx [file]
curl http://localhost:3000/
```



SKILLS SYSTEM

Downloaded Skills

Skill	Location	Status	Action
graph-memory	graph-memory/	Downloaded	Install
clawtrol	clawtrol/	Code Ready	Integrate
mnemon	mnemon/	Downloaded	Add to memory
clawbridge	clawbridge/	Downloaded	Mobile dashboard
agent-orchestrator	agent-orchestrator/	Patterns Ready	Study

Rule: "Downloaded but not used" = WASTED TIME

How to USE Skills:

1. Skill downloaded → READ its README/CLAUDE.md
2. Find patterns → COPY to our project
3. Test the pattern → Make sure it works
4. If works → INTEGRATE into engine
5. If fails → FIX before moving on

TESTING PROTOCOL

Before Claiming "Done":

- ❌ WRONG: "Ho gaya, code likh diya"
- ✅ RIGHT: "Code likha, test run kiya, output dekho..."

Test Output Template:

- ✅ [Feature] - WORKING
Output: [actual result]
Model Used: [which API]
Tokens: [if tracking]

Verification Checklist:

- ☐ Code runs without errors
- ☐ Output matches expectation
- ☐ Browser/App test done
- ☐ Git committed

API KEYS

Current Keys

Gemini API

```
Status: ❌ BLOCKED (403 Error)
Keys in .env:
- GEMINI_API_KEY=AIZA5yCIoT8G0SCBgDJAVhsdCgGrIciFF8rFvwM
- AI_INTEGRATIONS_GEMINI_API_KEY=AIZA5yCIoT8G0SCBgDJAVhsdCgGrIciFF8rFvwM

SOLUTION: Get new key from https://aistudio.google.com/apikey
```

Model Priority (Cost Optimization)

Priority	Model	Cost
1	Gemini 2.5 Flash	FREE
2	gemini-2.0-flash	FREE
3	OpenRouter	~\$0.001/1K
4	Claude Haiku	~\$0.001/1K

COMMANDS REFERENCE

Quick Commands

```
# Read rules
cat RULES.md

# Load memory
cat .claude/memory.json

# Test Gemini
curl -X POST "https://generativelanguage.googleapis.com/..."

# Start Backend
cd crm-official && npm run server

# Start CRM
cd crm && npx serve -p 3000

# Commit
git add . && git commit -m "message" && git push
```

CRM Commands

```
# Start 3D Industrial CRM
cd crm && npx serve -p 3000

# Open: http://localhost:3000
```

Backend Commands

```
# Start server
cd crm-official && npm run server

# Build web
cd crm-official && npm run build
```

```
# Mobile build
cd crm-official/mobile && eas build --platform android --profile preview
```



PLAY STORE GUIDE

Current Build

File: crm-official/android-release.aab
Size: 35MB
Date: March 29, 2026
Status: Ready to upload

Play Console Link

<https://play.google.com/console/u/2/developers/5557389597984591128/app/4976180857303773796>

Upload Steps (Beta Testing)

- 1. Open Play Console link above
- 2. Go to "Testing" tab
- 3. Select "Closed Testing" or "Open Testing"
- 4. Click "Create Release"
- 5. Upload android-release.aab
- 6. Add release notes
- 7. Click "Save & Publish"

Package Info

App Name: SAI RoloTech CRM
Package: com.vipinjangra.crmmobile
Version: 1.1.0



TROUBLESHOOTING

Common Errors & Solutions

Error	Solution
"memory.json not loaded"	cat .claude/memory.json
"rules not found"	cat RULES.md
"checkpoint failed"	cat .claude/CHECKPOINT.md
"API key not working"	Check RULES.md for API status

"Gemini 403"	Get new API key
"EAS permission error"	Check Expo account ownership

MEMORY FILES

Memory Structure

```
.claude/
├─ memory/
│   ├── MEMORY.md          # Main index
│   ├── user_*.md          # User info
│   ├── project_*.md       # Project info
│   ├── feedback_*.md      # User preferences
│   └── reference_*.md      # External references
└─ CHECKPOINT.md           # Validation key
```

Update Memory After:

- Completing a task
- Finding issues
- User preferences
- API status changes

QUICK START CHECKLIST

```
❑ Read RULES.md
❑ Test Gemini API
❑ Check backend server
❑ Understand project structure
❑ Execute task
❑ Test code
❑ Update memory
❑ Commit to git
```

IMPORTANT REMINDERS

NEVER FORGET

1. Before ANY task → Read RULES.md
2. After coding → Test immediately
3. After testing → Show output to user
4. Every 5 steps → Commit + Push


```
|| 5. API fail → Switch model, don't stop || | |
|| 6. Downloaded skill → USE it, not keep it ||
|| 7. Cheap model first → Gemini (free) > OpenRouter > Claude ||
|| || ||
|| ❌ "Ho gaya" → ✅ "Test kiya, output mila, done" ||
|| || ||
```

Document Info

- **Created:** 2026-04-14
- **Version:** 1.0
- **Last Updated:** 2026-04-14
- **Author:** Claude Code (Opus 4.6)

End of Document